

Name:

BU Id:

DS 120 Fall'25

Midterm 2 practice.

1. Let $V = \{(x, y) \in \mathbb{R}^2 : x - y = 1\}$ be a subset of $\mathbb{R}^2$ with standard vector addition and scalar multiplication. Is this a valid vector space?

Hint: Check for presence of 0 vector and closure under standard vector addition and scalar multiplication.

2. Let

$$\vec{v}_1 = \{1,1,0\}$$

$$\vec{v}_2 = \{0,1,1\}$$

$$\vec{v}_3 = \{1,2,1\}$$

Is $\vec{v}_3$ in $\text{span}\{\vec{v}_1, \vec{v}_2\}$? If yes, find coefficients.

3. Let $\mathbf{u}_1 = (1,0,1)$, $\mathbf{u}_2 = (0,1,1)$, $\mathbf{u}_3 = (1,1,2)$, $\mathbf{b} = (2,3,5)$.

Check if $\mathbf{b}$ is in $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$.

4. Let

$$S = \{(1,0,0), (0,1,0), (1,1,0)\}.$$

Does S generate all of $\mathbb{R}^3$?

5. Let

$$S = \{(1,0,0), (0,1,0), (1,1,0), (0,0,1)\}$$

be a spanning set of $\mathbb{R}^3$.

Find a **basis** by removing redundant vectors.

6. If $\det(A) = 0$, can you find A^{-1} ?

If not, explain **why** using reasoning.

7. If a 3×3 matrix A has $\det(A) = 0$, what happens if we replace one row with the sum of the other two?

8. Project $\mathbf{w} = (1,2,3)$ onto the line spanned by $\mathbf{a} = (1,0,1)$.

9. Model:

$$y = \beta_1 x_1 + \beta_2 x_2$$

Data:

Obs	x_1	x_2	y
1	3	5	10
2	1	2	5
3	4	6	12

Find the coefficient matrix

10. Given,

$$f(x, y) = x^2 + y^2 + xy$$

Is (0,0) a saddle point?

11. Given, Potential energy of a particle is

$$U(x, y, z) = 2x^2 + y^2 + 4z^2.$$

Compute the **direction** in which the potential increases most rapidly at point $(1, -2, 1)$, and find the **rate** of increase in the direction of vector $\mathbf{v} = (1, 1, 0)$.

12. Find the **area under** the curve $y = x^2$ from $x = 0$ to $x = 3$